

InvoraLite — Product Quality Review

Review date: 13 July 2026 (late evening) | **Exported:** 13|07|2026 | **Version:** 1.0.0 | **Stack:** Tauri 2 + React 19 + SQLite | **Includes:** Testing & Quality Checkup

Executive Summary

Overall Rating: 9.1 / 10

InvoraLite is a mature v1.0 offline desktop inventory and retail application for a single Windows store. Coverage includes sales, purchases, Rate Master, **20-type Offers**, Quotations, credit, GST invoices, analytics, accounting, two-stage returns, purchase history, supplier ledgers, and a **loyalty points MVP**.

This late-evening recheck closes several prior quality gaps: **GitHub Actions CI, Playwright E2E smoke scaffolding, vault unclean-exit / logout seal hardening**, and customer **loyalty earn/redeem**.

Automated checkup (this run): **169 / 169 Vitest, 30 / 30 Rust, 2 / 2 Playwright smoke**, TypeScript clean. Ready for trusted single-store deployment. Before wide public distribution: rotate license secrets and code-sign the installer (relational schema and auto-update remain future work).

Rating Breakdown

Area	Score	Notes
Feature breadth	9.4/10	Returns + loyalty points MVP + full pricing hub
UX / polish	8.4/10	Return modals; loyalty redeem on New Sale; People points display
Business logic	9.0/10	Offers, quote→sale, returns, loyalty earn/redeem/reverse
Security	8.5/10	Vault refresh on unclean exit; seal on logout; license secret still ship-blocker
Data / reliability	7.8/10	Atomic batches; JSON-in-SQLite still the scale ceiling
Testing & quality	8.3/10	CI + 169 Vitest + 30 Rust + Playwright smoke; no deep Tauri E2E yet
Deployment	7.5/10	NSIS; CI ready; no code signing / auto-update

Technology Stack

Layer	Technology
Desktop shell	Tauri 2, WebView2
Frontend	React 19, TypeScript 5.6 (strict), Vite 6, Tailwind 4
Charts	Recharts 2.15
Backend	Rust 2021 — src-tauri/src/
Database	SQLite via rusqlite (WAL); domain JSON in app_storage
Encryption	AES-256-GCM vault, Argon2id backups, Windows DPAPI key file
CI / E2E	GitHub Actions (windows-latest); Playwright Chromium smoke vs Vite
IPC	storage_get/set/set_many, seal_at_rest, ensure_open, refresh_vault, audit, auth, license, backup

Architecture

```
React (pages, components, lib/data.ts, returns.ts, loyalty.ts, cartOffers.ts)
  → storage.ts (username + role as StorageContext)
    → Tauri commands (storage_set_many; database_seal_at_rest / ensure_open)
      → DatabaseManager (transactions · vault seal/unseal · RBAC · audit)
```

Feature Completeness

Module	Status	Notes
Products / Sales / Purchases	Complete	Atomic stock+sale; purchase history
Sales / Purchase Returns	Complete	Two-stage linked documents
Loyalty points	MVP	Earn on sale; redeem at till; reverse on return; LOYALTY offer rates
Rate Master / Offers / Quotations	Complete	20 offer types live; quote convert via submitSale
Users / RBAC	Aligned	Manager in Rust; return keys for roles
CI / E2E smoke	Shipped	.github/workflows/ci.yml + Playwright scaffolding

Gaps closed this programme (13|07|2026)

Prior gap	Resolution
No CI workflow	Fixed — GitHub Actions: tsc, Vitest, cargo, Playwright
No E2E suite	Scaffolded — Playwright smoke vs Vite (Tauri deep E2E later)
Unclean exit / plaintext vault drift	Hardened — refresh vault; seal on logout/timeout
Loyalty group-only (no points)	MVP — balance earn/redeem/reverse
Two-stage returns	Shipped (earlier this evening)

Remaining gaps

- **License HMAC secret** still a placeholder — rotate before production keys
- JSON blob domain model (fine for single PC; limits multi-cashier scale)
- Windows NSIS only; no code signing / auto-update
- Runtime DB remains plaintext *while a session is open* (SQLCipher not adopted)
- Loyalty: no full append-only ledger / expiry / points reports yet
- Playwright covers web smoke only — not full desktop WebView2 paths

Security Posture

Control	Status
Vault at rest + DPAPI key	Strong
Unclean-exit vault refresh + logout seal	Hardened
Argon2id, lockout, session timeout	Strong
UI ↔ Rust RBAC (incl. Manager)	Aligned
License secret in source	High — rotate before distribution
No code signing	Medium (SmartScreen)

Testing & Quality Checkup

Run date: 13 July 2026 (late evening) | Result: **ALL PASSED**

Automated test results

Suite	Command	Cases	Result
Frontend (Vitest)	npx vitest run	26 files · 169 tests	169 passed · 0 failed
Backend (Rust)	cargo test	30 tests	30 passed · 0 failed
E2E smoke (Playwright)	npm run test:e2e	2 tests	2 passed · 0 failed
TypeScript	tsc --noEmit	—	Clean

Coverage highlights

Area	Coverage
Rate Master / Offers / cart / quotations	Strong unit coverage
Returns + supplier ledger debit notes	returns.test.ts + data.test.ts
Loyalty earn / redeem / reverse	loyalty.test.ts
Vault unclean-exit refresh	Rust encryption_key test
CI pipeline	.github/workflows/ci.yml
Deep Tauri UI E2E	Not yet (scaffolding only)

Manual smoke checklist (recommended)

1. Login → New Sale with customer → confirm loyalty points increase on Customers
2. Redeem points on next sale; balance decreases; invoice discount applied
3. Invoice → Sales Return → Purchase Return (supplier-liable)
4. Logout — confirm vault sealed (no live invora.db until next open)
5. Quotation convert; Manager write; encrypted backup export/restore

Recommended Improvements

Before wider distribution

1. Replace LICENSE_SECRET; remove any hardcoded pack passwords
2. Code-sign the Windows installer

Medium term

3. Expand Playwright to seeded sale/return flows; optional Tauri WebDriver
4. Loyalty append-only ledger + expiry + reports
5. Auto-update channel after signing + update host
6. Relational migration only if multi-cashier concurrency becomes real

Changes Since Prior Review (evening → late evening 13|07)

Item	Evening	Now
Overall rating	8.9 / 10	9.1 / 10
CI	None	GitHub Actions
E2E	None	Playwright smoke (2)
Vault unclean / logout	Documented risk	Hardened
Loyalty points	Group discount only	MVP earn/redeem
Vitest	165	169
Rust	29	30

Use-Case Fit

Scenario	Fit
Owner's shop, trusted staff, offers/quotes/returns/loyalty	9.4/10
Few clients, supervised rollout (after license rotation)	8.8/10
Public license distribution	7.2/10 — needs secret rotation + signing
Multi-store / enterprise	4/10 — single-PC JSON model

Top Strengths

1. Full retail loop — pricing hub → sale → invoice → two-stage returns
2. Hardened persistence — atomic batches + vault seal/refresh + logout seal
3. Automated quality gate — CI with unit + smoke E2E
4. Loyalty MVP wired through sale and return without a separate ledger yet
5. RBAC aligned across UI and Rust

Top 3 Next Steps

1. **Production license hardening** — rotate secrets
2. **Code-sign installer** — reduce SmartScreen friction
3. **Expand E2E** — seeded sale / return / loyalty paths

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Product Quality Review & Testing Checkup · EDP IT Department

See also docs/DATABASE_ARCHITECTURE.md and docs/InvoraLite_User_Manual.pdf

13|07|2026